PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Comparison of Hydrus and iStent microinvasive glaucoma surgery implants in combination with phacoemulsification for treatment of open-angle glaucoma: systematic review and network meta-analysis
AUTHORS	Hu, Rongrong; Guo, Dongyu; Hong, Nan; Xuan, Xiuyuan; Wang, Xiaoyu

VERSION 1 – REVIEW

REVIEWER	Konopińska, Joanna Medical University of Bialystok
REVIEW RETURNED	20-May-2021

GENERAL COMMENTS	This is a very interesting study, hovewer my main concern is that, the in most cases the historical data vere analyzed. The authors included only one RTC from 2019, while the most other references are older than 2015. During the serching for pub med I have found some other, more recent studies that you did not included into your meta-analysis for example: Konopińska J, Kozera M, Kraśnicki P, Mariak Z, Rękas M. The Effectiveness of First-Generation iStent Microbypass Implantation Depends on Initial Intraocular Pressure: 24-Month Follow-Up-Prospective Clinical Trial. J Ophthalmol. 2020 Jun 23;2020:8164703. doi: 10.1155/2020/8164703. PMID: 32676205; PMCID: PMC7330650. Kozera M, Konopińska J, Mariak Z, Rękas M. Effectiveness of iStent Trabecular Microbypass System Combined with Phacoemulsification versus Phacoemulsification Alone in Patients with Glaucoma and Cataract Depending on the Initial Intraocular Pressure. Ophthalmic Res. 2021;64(2):327-336. doi: 10.1159/000511456. Epub 2020 Sep 9. PMID: 32906138. Katz LJ, Erb C, Carceller Guillamet A, Fea AM, Voskanyan L, Giamporcaro JE, Hornbeak DM. Long-term titrated IOP control with one, two, or three trabecular micro-bypass stents in open-angle glaucoma subjects on topical hypotensive medication: 42-month outcomes. Clin Ophthalmol. 2018 Jan 31;12:255-262. doi: 10.2147/OPTH.S152268. PMID: 29440867; PMCID: PMC5798569. My second major concern is that the authors did not state which kind of iStent was analyzed (first generation, second?). And the discussion should include paragraph, if there were any diferennces between these two kinds of iStents in the subject of effectivenes and the rate of complications.

REVIEWER	Akkach, Sarmad Columbia University
REVIEW RETURNED	03-Oct-2021

GENERAL COMMENTS	Thank you for submission to BMJ and for giving me the pleasure of reviewing this manuscript. It is a clinically relevant study with useful findings. Generally, it is cogent and well-written. My main query is why you chose to present the findings using rank probability analysis? The justification and clinical utility of this decision was unclear to me. Importantly, I think it would be more useful to the clinician if IOP reduction (1 istent, 2 istent, Hydrus and control) from the meta analysis was reported with routine statistical analysis. This would allow you to ultimately present the key findings (IOP reduction and adverse events) in a more simplistic graph then is presented in Figure 4. If this study is intended to aid the clinician in the decision making paradigm, I believe a clearer presentation of IOP reduction would be more beneficial. In table 5. Clarification is needed with regard to NA and NR. If NR indicates 'not reported' I assume NA means there were 0 complications? If this is the case, it would be more intuitive to simply state 0 in Table 2.

REVIEWER	Murdoch, Ian UCL, Institute of Ophthalmology
REVIEW RETURNED	22-Dec-2021

GENERAL COMMENTS	The authors report a meta-analysis of 6 studies reported as RCTs comparing phacoemulsification alone with insertion of either iStent or Hydrus intra-canalicular implants as the time of phacoemulsification. They report a larger effect on IOP lowering for both implants compared with phacoemulsification alone. The methods and analysis are well described. I have some comments on the current presentation. 1 In my opinion meta-analysis is best performed using the original data since this allows much improved analysis rather than simply weighting summary findings by size of study. If this is not achieved then the usual ‘fudge’ to get around this is a forest plot. Such a plot should be included together with some attention to the limitations and the advice on the Cochran website copied below "There are 3 main things we need to assess when reading a meta-analysis: 1. Heterogeneity. The differences in the results, methodology or study populations used in the included studies. 2. The pooled result. The overall combined result derived from combining (‘pooling’) the individual studies. 3. Publication bias. Although the intent of a meta-analysis is to find and assess all the relevant studies meeting the inclusion criteria, this mission is not always possible. Some studies can be missed because they are not written in English, or because they show non-significant results (so they have a lower chance of being published). A forest plot does a great job in illustrating the first two of these (heterogeneity and the pooled result). However, it cannot display potential publication bias to readers. A funnel plot can do that instead." 2 A meta-analysis might be reasonably expected to include comments on the variability of the populations in each study such as ethnicity, severity of glaucoma, copathology etc.

	3 The clinical importance of the findings need discussion in context of costs, availability of equipment and skills and relevance to the case/context of the additional pressure lowering achieved.

VERSION 1 – AUTHOR RESPONSE

RESPONSE TO REVIEWERS

Authors: We are thankful for the insightful comments provided by the Editor and reviewers that have helped us to refine our study and have also pushed us to improve the presentation of our work. We carefully address each comment below.

1. Comments from the Editor:-

Please go through the PRISMA extension for abstracts (http://www.prisma-statement.org/Extensions/Abstracts.aspx) and check that items 1-10 are reported in your abstract.

Authors: The abstract has been revised based on the PRISMA Abstracts Checklist. We specified the inclusion criteria, the method used to assess risk of bias, and the main limitation of the evidence included in the review.

1. You have indicated that you have not included patients or the public in your study design because it is a systematic review. Please note that patients and the public can be involved in all study designs, including systematic review. We ask that you revise your PPI statement to be in line with our Instructions for Authors at https://bmjopen.bmj.com/pages/authors/#reporting_patient_and_public_involvement_in_research. If patients and or public were not involved, please state this.

Authors: We have revised the PPI statement according to the instructions. Please see lines 102-107.

1. Please include an example of the full search strategy for one database as a supplementary file.

Authors: We include the full search strategy used in PubMed in the Appendix file.

1. We felt the manuscript gives the impression it was a standard pair-wise meta-analysis comparing Hydrus with istent, but it is a network meta-analysis comparing four diferent treaments: Hydrus implantation with phacoemulsification, 1-iStent implantation with phacoemulsification, 2-iStent implantation with phacoemulsification, and phacoemulsification alone. Please explain what is the difference between 1-iStent and 2-iStent?

Authors: We have added a separate paragraph to detailedly discuss the difference between 1-iStent and 2-iStent in treatment efficacy (lines 376-392). Taken together, Hydrus might provide slightly better efficacy than 2-iStent, and 2-iStent might be slightly better than 1-iStent. As the overall safety profiles of 1-iStent, 2-iStent and Hydrus were all favorable in included RCTs, we discussed the complications of the three groups together in the same paragraph (lines 394-408).

1. We note this includes only 6 studies, which is very thin for a network meta-analysis. The efficacy estimates have very wide confidence intervals so we felt you need to significantly tone down the findings and acknowledge far more uncertainty.

Authors: We significantly tone down and acknowledge the uncertainty throughout the manuscript (Abstract line 43-44, Discussion, and Conclusion 460-464).

Despite the increasing use of MIGS in clinical practice, high-quality data to directly compare these devices remain limited. Using previously published RCT data, we hope our indirect comparison findings based on the network meta-analysis could provide some evidence for clinical considerations. However, our finding does have uncertainty due to the limited included data. Further research is needed to investigate whether our findings are robust.

1. Formatting Amendments (where applicable):

(1). Required Supplementary format:
- Please re-upload your Supplementary files in PDF format.

Authors: All supplementary files have been uploaded in PDF format.

(2). No ethics approval in main doc:
- We require every research article submitted to include a statement that the study obtained ethics approval (or a statement that it was not required and why), including the name of the ethics committee(s) or institutional review board(s), the number/ID of the approval(s), and a statement that participants gave informed consent before taking part.

Authors: we added relevant information related to the ethics approval in lines 484-486.

1. Reviewer: 1

Dr. Joanna Konopińska, Medical University of Bialystok
Comments to the Author:
This is a very interesting study, hovewer my main concern is that, the in most cases the historical data vere analyzed. The authors included only one RTC from 2019, while the most other references are older than 2015. During the serching for pub med I have found some other, more recent studies that you did not included into your meta-analysis for example:
Konopińska J, Kozera M, Kraśnicki P, Mariak Z, Rękas M. The Effectiveness of First-Generation iStent Microbypass Implantation Depends on Initial Intraocular Pressure: 24-Month Follow-Up-Prospective Clinical Trial. J Ophthalmol. 2020 Jun 23;2020:8164703. doi: 10.1155/2020/8164703. PMID: 32676205; PMCID: PMC7330650.
Kozera M, Konopińska J, Mariak Z, Rękas M. Effectiveness of iStent Trabecular Microbypass System Combined with Phacoemulsification versus Phacoemulsification Alone in Patients with Glaucoma and Cataract Depending on the Initial Intraocular Pressure. Ophthalmic Res. 2021;64(2):327-336. doi: 10.1159/000511456. Epub 2020 Sep 9. PMID: 32906138.

Katz LJ, Erb C, Carceller Guillamet A, Fea AM, Voskanyan L, Giamporcaro JE, Hornbeak DM. Long-term titrated IOP control with one, two, or three trabecular micro-bypass stents in open-angle glaucoma subjects on topical hypotensive medication: 42-month outcomes. Clin Ophthalmol. 2018 Jan 31;12:255-262. doi: 10.2147/OPTH.S152268. PMID: 29440867; PMCID: PMC5798569.

Authors: Thank you for recommending these relevant studies. In the present study, we included only RCTs which usually minimize confounding and allocation bias, and serve as a valid methodology for health care decisions.[1] We carefully read all the relevant articles to determine if they contained data that were applicable to our study. The inclusion and exclusion criteria were explained in "Methods" section in detail (lines 122-130). Finally, we included 6 RCTs, Fea et al. (2010), Craven et al. (2012), Fernandez-Barrientos et al. (2010), Samuelson et al. (2019), Pfeiffer et al. (2015), and Samuelson et al. (2019).

Both Dr. Konopińska's and Dr. Kozera's studies investigated the treatment efficacy of iStent implantation in terms of baseline IOP levels. The studies have great practical value as their results explained why the efficacy of the iStent may be limited in eyes with higher baseline IOP (e.g. baseline IOP >26 mmHg). Both studies divided participants into groups based on the baseline IOP level, which had heterogeneity in the study methodology compared with our included RCTs. Both studies analyzed data in depth, however, Dr. Konopińska's study did not showphacoemulsification alone as control, and Dr. Kozera's study did not report the number or proportion of medication-free patients after surgery, which is one of our outcome measures.

Dr. Katz's study investigated the treatment outcomes of one, two, or three iStents as a standalone procedure, which is out of our study scope.

1. My second major concern is that the authors did not state which kind of iStent was analyzed (first generation, second?). And the discussion should include paragraph, if there were any diferennces between these two kinds of iStents in the subject of effectivenes and the rate of complications.

Authors: For 1-iStent group, we only included first-generation iStent data. For the 2-iStent group, we merged the data analyzing 2 first-generation stents (Fernandez-Barrientos et al. 2010)[2] and second-generation iStent *inject*, which is preloaded with 2 stents (Samuelson et al. 2019).[3] Relevant statements have been added in the Introduction (lines 89-92) and table 1. We also specified which kind of iStent was investigated by the cited literatures in the Discussion section.

During the preparation of the present study, we searched the literature for studies conducted to compare 2 first-generation stents and second-generation iStent inject. However, few reports were found. In vitro perfusion model study did not demonstrate significant difference in outflow facility increase between the 2 stents.[4] A subgroup analysis by Salimi et al (Am J Ophthalmol 2021;224:310–320)[5] shows no significant difference was found between 2 first-generation stents and second-generation iStent inject in terms of IOP reduction and medication use in eyes with primary angle closure glaucoma. Though the difference between these 2 stents is uncertain, it might be a limitation of our study to merge the data and we address this limitation in lines 447-454.

For the differences between a single first-generation stent and multiple first-generation stents, and between a single first-generation stent and double second-generation iStent inject, we have added detailed discussion based on the present network meta-analysis and evidence from literature (lines 376-392).

"The theoretical basis for incremental efficacy with implantation of multiple trabecular stents was established previously based on in vitro perfusion models.[37-39] Several clinical studies have shown that use of additional first-generation iStent may result in greater IOP reduction with reduced medication use versus a single implant.[40-42] One study by Hooshmand et al[18] did

not demonstrate a statistically significant difference between a single iStent and double iStent inject during one-year follow-up, and earlier recommencement of medications for optimal IOP control was reported with the iStent inject. In the present study, the majority of study participants of the 2-iStent group were implanted with the iStent inject. Though there was no significant difference between the 1-iStent group and the 2-iStent group across our outcome measures, the 2-iStent group showed statistically significant reduction in absolute values of IOPR versus phacoemulsification alone, while the 1-iStent group not. Furthermore, the rank probability analysis shows the 2-iStent implantation may achieve complete success at a higher probability than the 1-iStent implantation (0.44 vs. 0.16, respectively). Larger population size and longer term follow-up might be needed to obtain more definitive results for the efficacy comparison between a single first-generation iStent and second-generation iStent inject."

1. Reviewer: 2

Dr. Sarmad Akkach, Columbia University
Comments to the Author:
Thank you for submission to BMJ and for giving me the pleasure of reviewing this manuscript. It is a clinically relevant study with useful findings. Generally, it is cogent and well-written.

My main query is why you chose to present the findings using rank probability analysis? The justification and clinical utility of this decision was unclear to me. Importantly, I think it would be more useful to the clinician if IOP reduction (1 istent, 2 istent, Hydrus and control) from the meta analysis was reported with routine statistical analysis. This would allow you to ultimately present the key findings (IOP reduction and adverse events) in a more simplistic graph then is presented in Figure 4. If this study is intended to aid the clinician in the decision making paradigm, I believe a clearer presentation of IOP reduction would be more beneficial.

Authors: We agree that a clearer presentation of treatment efficacy would be more beneficial to aid clinicians in the decision making paradigm. Network meta-analysis was developed to assess the relative efficacy of several treatments by synthesizing evidence across a network of randomized trials.[6] It allows the estimation of the probability that each treatment is the best for each outcome, which may be a straightforward way to present network results as clinicians usually want to know the preferential order of treatments for an average patient. This methodology has also been used by other studies.[7,8] For clarity, we have revised the Figure 4. As shown in the updated Figure 4, Hydrus has the higher probability of being ranked the best (Rank 1) compared with either 1-iStent or 2-iStent in IOP reduction.

Figure 4 (revised)

1. In table 5. Clarification is needed with regard to NA and NR. If NR indicates 'not reported' I assume NA means there were 0 complications? If this is the case, it would be more intuitive to simply state 0 in Table 2.

Authors: NA, which means “not applicable”, was used to state there was no device-relevant complication (malposition or obstruction) in the control groups treated with phacoemulsification alone. We added supplementary information in the table legend (lines 313-314).

1. Reviewer: 3

Dr. Ian Murdoch, UCL
Comments to the Author:
The authors report a meta-analysis of 6 studies reported as RCTs comparing phacoemulsification alone with insertion of either iStent or Hydrus intra-canalicular implants as the time of phacoemulsification. They report a larger effect on IOP lowering for both implants compared with phacoemulsification alone. The methods and analysis are well described. I have some comments on the current presentation.

1 In my opinion meta-analysis is best performed using the original data since this allows much improved analysis rather than simply weighting summary findings by size of study. If this is not achieved then the usual ‘fudge’ to get around this is a forest plot. Such a plot should be included together with some attention to the limitations and the advice on the Cochran website copied below
"There are 3 main things we need to assess when reading a meta-analysis:
1. Heterogeneity. The differences in the results, methodology or study populations used in the included studies.
2. The pooled result. The overall combined result derived from combining (‘pooling’) the individual studies.
3. Publication bias. Although the intent of a meta-analysis is to find and assess all the relevant studies meeting the inclusion criteria, this mission is not always possible. Some studies can be missed because they are not written in English, or because they show non-significant results (so they have a lower chance of being published).
A forest plot does a great job in illustrating the first two of these (heterogeneity and the pooled result). However, it cannot display potential publication bias to readers. A funnel plot can do that instead."

Authors: We now include the supplemental figures 1-3 to show the forest plots and relevant introduction of the statistical methodology has been addressed in lines 185-190. As shown in supplemental figures 1-3, the overall heterogeneity is not significant among included RCTs, except the 1-iStent group data for the proportions of complete success (I^2 = 72.3%). As we condcted the bayesian network meta-analysis, forest plots based on traditional pair-wise meta-analysis might not be feasible well.[8] Also due to the limited numbers of included studies, funnel plot would be hard to illustrate heterogeneity and potential publication bias. Alternatively, we use the 6-item modified Jadad scale to comprehensively evaluate the methodological quality, which includes the method of randomization, blinding, withdrawals and drop-outs, inclusion/exclusion criteria, adverse events, and the statistical analysis.

1. 2 A meta-analysis might be reasonably expected to include comments on the variability of the populations in each study such as ethnicity, severity of glaucoma, copathology etc.

Authors: we include a new table (Table 2) to present the baseline characteristics of study populations of included RCTs. Relevant comments about the baseline variability of included populations have been added in lines 410-422.

“The baseline characteristics of study populations were overall similar in included RCTs. A common preoperative diagnosis was OAG, including 3 studies only with primary open-angle glaucoma,[9,13,15] one study with ocular hypertension (11.8%),[12] one study

with pseudoexfoliative glaucoma (6%) and pigmentary glaucoma (3%),[11] and one study with pseudoexfoliative glaucoma (10%).[14] Angle-closure, traumatic, uveitic, or neovascular glaucoma, or history of incisional glaucoma surgery were all excluded. Baseline visual field damage was comparable in included RCTs, except one Hydrus study with greater mean deviation and pattern standard deviation.[14] Real-world data show that the Hydrus have been used in patients with more advanced disease versus the iStent inject, which are in line with the present included RCT.[36] As shown in table 2, the majority of study populations were Caucasian. There was no significant difference in age and sex among groups and these demographics characteristics were well matched for the treatment and control groups in each study."

1. 3 The clinical importance of the findings need discussion in context of costs, availability of equipment and skills and relevance to the case/context of the additional pressure lowering achieved.

Authors: we add relevant discussion in lines 424-438.

"The surgical techniques of the Hydrus and the iStent are similar which both require a clear view of the angle structure, and the devices are introduced into the anterior chamber through a clear corneal incision and implanted through the trabecular meshwork.[34] Despite the increasing use of MIGS in clinical practice, data to compare the cost-effectiveness of these devices remain limited.[43] Separate studies have shown either the Hydrus[44] or the iStent inject[45,46] combined with phacoemulsification is cost-effective for patients with mild to moderate OAG versus phacoemulsification alone. A recent study by Sood et al[47] compared the cost-effectiveness of the Hydrus and the iStent inject in combination with phacoemulsification for patients with mild to moderate primary open-angle glaucoma. Both devices were not cost-effective versus phacoemulsification alone over the first 2 years due to the initial costs of the surgeries, however, these devices were demonstrated to be cost-effective treatment options over a lifetime horizon based on the simulation analysis. In comparison, the Hydrus was slightly more cost-effective than the iStent inject, and the authors concluded it was likely due to the slightly better treatment efficacy associated with the Hydrus versus the iStent inject at the identical costs."

References

1. Sibbald B, Roland M. Understanding controlled trials. Why are randomised controlled trials important? *BMJ* 1998;316:201.
2. Fernandez-Barrientos Y, Garcia-Feijoo J, Martinez-de-la-Casa JM, et al. Fluorophotometric study of the effect of the glaukos trabecular microbypass stent on aqueous humor dynamics. *Invest Ophthalmol Vis Sci* 2010;51:3327-32.
3. Samuelson TW, Sarkisian SR, Jr., Lubeck DM, et al. Prospective, Randomized, Controlled Pivotal Trial of an Ab Interno Implanted Trabecular Micro-Bypass in Primary Open-Angle Glaucoma and Cataract: Two-Year Results. *Ophthalmology* 2019;126:811-21.
4. Toris CB, Pattabiraman PP, Tye G, et al. Outflow Facility Effects of 3 Schlemm's Canal Microinvasive Glaucoma Surgery Devices. *Ophthalmol Glaucoma* 2020;3:114-21.
5. Salimi A, Abu-Nada M, Harasymowycz P. Matched Cohort Study of Cataract Surgery With and Without Trabecular Microbypass Stent Implantation in Primary Angle-Closure Glaucoma. *Am J Ophthalmol* 2021;224:310-20.
6. Cipriani A, Higgins JP, Geddes JR, et al. Conceptual and technical challenges in network meta-analysis. *Ann Intern Med* 2013;159:130-7.
7. Wu HY, Huang JW, Lin HJ, et al. Comparative effectiveness of renin-angiotensin system blockers and other antihypertensive drugs in patients with diabetes: systematic review and bayesian network meta-analysis. *BMJ* 2013;347:f6008.
8. Liao WC, Chien KL, Lin YL, et al. Adjuvant treatments for resected pancreatic adenocarcinoma: a systematic review and network meta-analysis. *Lancet Oncol* 2013;14:1095-103.

VERSION 2 – REVIEW

REVIEWER	Konopińska, Joanna Medical University of Bialystok
REVIEW RETURNED	05-Apr-2022

GENERAL COMMENTS	According to the point 16B PRISMA statement the authors should "Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded ", so the authors should cite papers of Dr. Konopińska's and Dr. Kozera's who studies investigated the treatment efficacy of iStent implantation in terms of baseline IOP levels anyway.

REVIEWER	Akkach, Sarmad Columbia University
REVIEW RETURNED	28-Apr-2022

GENERAL COMMENTS	Thank you for amending figure 4 and table 5, they are presented much clearer. I am happy for this to be published now. Well done.

VERSION 2 – AUTHOR RESPONSE

VERSION 3 – REVIEW

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

VERSION 3 – AUTHOR RESPONSE

VERSION 4 – REVIEW

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

VERSION 4 – AUTHOR RESPONSE